Programme Code: TU757, TU758, TU860, TU863
Module Code: COMP H2026
CRN:17102

TECHNOLOGICAL UNIVERSITY DUBLIN

BLANCHARDSTOWN CAMPUS

TU757 Bachelor of Science in Computing in Information Technology
TU758 Bachelor of Science in Computing in Digital Forensics & Cyber Security
TU860 Bachelor of Science (Honours) in Computing
TU863 Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security

Year 2

SEMESTER 1
EXAMINATIONS 2023/24

Thursday 4th January 2024
15:30pm - 17:30pm

Information Technology Mathematics

Internal Examiner: Laura Keyes
External Examiner: Dr. Chris Meudec

Exam Duration: 2 Hours
Instructions:

- ***Question 1 is compulsory. Candidates should attempt ALL parts of Question 1 and any other TWO Questions.***
- ***This paper is worth 100 marks. Question 1 is worth 40 marks. The remaining questions are worth 30 marks each.***
- ***Remember to mark each number and section clearly and to list the questions which you have attempted in the correct order on the front of your exam script.***
- ***Show all you work.***

Question 1 (Compulsory) : All topics

Attempt ALL Parts

a) Given the following matrices:

$$A = \begin{bmatrix} -2 \\ 6 \\ -1 \end{bmatrix}, B = \begin{bmatrix} 3 & 0 \\ 1 & -2 \\ 5 & 2 \end{bmatrix} \text{ and } C = \begin{bmatrix} 5 & 1 & 3 \\ 3 & 2 & 7 \end{bmatrix}$$

i. What is the *size* of each matrix ***A, B*** and ***C***?
ii. Write down the value of the following elements: a_{21}, b_{22}, and c_{23}
iii. Find the matrix given by $2B^T - C$

(5 marks)

b) Find the *determinant* of

$$A = \begin{bmatrix} 1 & 7 & 5 \\ 3 & 0 & 1 \\ 6 & 3 & 2 \end{bmatrix}$$

(5 marks)

c) The following weight matrix shows the time taken in minutes to travel between 4 different towns represented as *A*, *B*, *C* and *D*.

	A	B	C	D
A	0	16	35	∞
B	16	0	20	18
C	35	20	0	34
D	∞	23	34	0

i. Draw the graph for the given weight matrix.
ii. How long does it take to travel from town *D* to town *B*?

(5 marks)

d) Evaluate t(2) and t(3) and t(4) for the following recursively defined sequence:

$$t(1) = 4$$
$$t(n) = t(n-1) + 2 \qquad (n>1)$$

(5 marks)

e) Using the data 3, 7, 2, 9, 8, 10 compute the

i. Mean
ii. Standard Deviation

(5 marks)

f) A new board game uses a 12-sided die. If the die is rolled once, what is the probability of getting:

i. an even number
ii. a number greater then 9
iii. an even number or a number greater than 9 **(5 marks)**

g) Calculate the gradient of the curve defined by $y = 6x^2 - 3x + 1$ at the point (2,1) **(5 marks)**

h) Evaluate the following *indefinite* integrals:

i. $\int(8x^3 + 3x^2 - 6x + 7).dx$
ii. $\int(x^{-2}).dx$ **(5 marks)**

Total: 40 marks

Question 2: Differentiation and Integration

a) If two variables x and y are related by the equation $y = 4x^2 - x + 2$, calculate the **average rate of change** of y with respect to x as x varies from 0 to 2. **(5 marks)**

b) Given the function $y = x^3+3x^2 - 9x + 5$ as shown in figure 2.1 below.

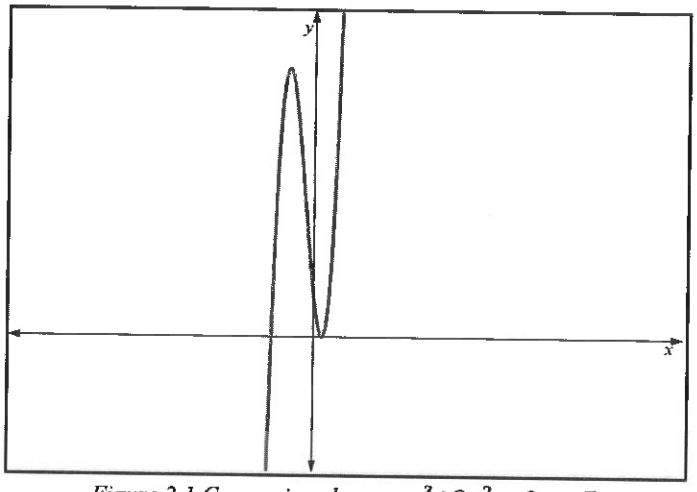


Figure 2.1 Curve given by $y = x^3+3x^2 - 9x + 5$

i. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. **(4 marks)**

ii. Using the second derivative test, find the coordinates of the *local maximum* and *local minimum* turning points of the function. **(6 marks)**

c) Differentiate the function $y = (3x^2 + x)(4x - 5)$ using the product rule. **(5 marks)**

d) The height h (in meters) of a ball at any time t seconds after the ball is hit is given by the equation $h(t) = -3t^2 + 24t$. Find:

i. The height of the ball 1.5 seconds after it is hit.
ii. The rate at which the ball is rising after 2 seconds.
iii. The time at which the ball reaches its maximum height. **(5 marks)**

e) Find the area enclosed by the curve $y = 2x^2 + 2x + 2$ for x between 1 and 4. **(5 marks)**

Total: 30 marks

Question 3: Statistics and Probability

a) A tutor surveyed their current class of 140 students about how they travel to and from college. 43 students responded to the survey. The following table 3.1 presents the data collected.

Table 3.1

Mode of transport	Number of students
Bus	21
Train	1
Car	16
Cycle	2
Walk	2
Other	1

i. Briefly discuss the collection and sourcing of data for statistical analysis under the following headings: *primary data*, *secondary data* and *sampling*. **(4 marks)**

ii. Is this primary or secondary data? Give reasons for your answer. **(1 mark)**

iii. State whether the data presented is *categorical* or *continuous*. Give reasons for your answer. **(1 mark)**

iv. Provide an appropriate graphical visualization of the given data. **(3 marks)**

b) The scatter plot below shows the amount of time a group of students spent studying for a math exam and the mark (%) they achieved in the exam.

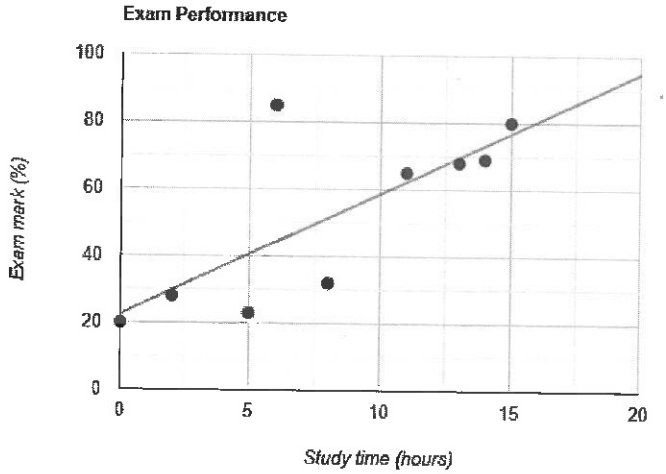


Figure 3.1

i. Describe the type of *correlation* shown in figure 3.1 and the relationship of time spent studying with the mark achieved in the exam. **(2 marks)**

ii. What is an *exception*? In your answer discuss any exception in the data presented above. **(2 marks)**

iii. Predict the exam mark of a student who spends 10 hours studying. **(1 mark)**

c) The following is the distribution of the time in minutes a sample of 43 students spend travelling to college.

Table 3.2

Time (minutes)	Frequency (No. of Students)
0 - 20	9
20 - 40	9
40 - 60	7
60 - 80	4
80 - 100	3
100 - 120	11

For the grouped data presented in Table 3.2:

i. Calculate the **Mean** of the grouped data. **(3 marks)**

ii. Calculate the **Standard Deviation** of the grouped data. **(5 marks)**

d) How many 4 letter codes can be constructed using the first 12 letters of the alphabet if no letter can be repeated? **(3 marks)**

e) The heights of a group of students can be modelled by a normal distribution with a mean of 165cm and a standard deviation of 4.5cm. Find the probability the height of a student selected at random will be less than 167cm. **(5 marks)**

Total: 30 marks

Question 4: Matrices, Graphs & Trees

a) Let $A = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 10 & 1 \\ 7 & 4 & 1 \end{bmatrix}$

i. Write down the *Identity Matrix* of A **(2 marks)**

ii. Evaluate $2B - AB$ **(5 marks)**

iii. Calculate the *determinant* and *inverse* of the matrix A **(5 marks)**

b) Consider the following graph $G = (V, E)$ (figure 4.1), starting at node A, in what order will the nodes be visited using

i. Breadth First Search
ii. Depth First Search

Assume, if there is a choice between nodes, the algorithm will choose alphabetically.

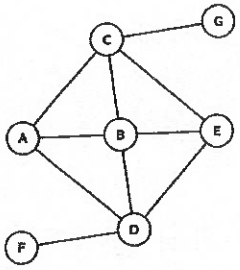


Figure 4.1 Graph G

(10 marks)

c) For the weighted graph in Figure 4.2 below, use *Prim's algorithm*, beginning with vertex A, to find a *Minimal Spanning Tree* and give its total *weight*.

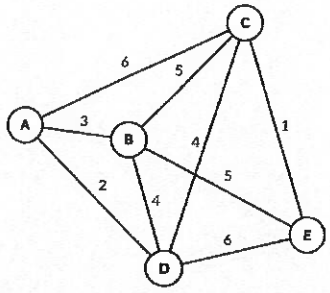


Figure 4.2 Weighted Graph

(8 marks)

Total:30 Marks

Formulae

Matrix Determinant

$\det A = ad - bc$ (2x2 matrix)

$\det A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$ (3x3 matrix using row 1)

Matrix Inverse

$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (2x2 matrix)

$A^{-1} = \frac{1}{\det A}C^T$, where C is the matrix of Cofactors (3x3 matrix)

General Probability:

If two events A and B are mutually exclusive, then the following law holds:

$P[A \text{ or } B] = P[A] + P[B]$

If the two events are not mutually exclusive, the law above is broadened so that the probability of A or B occurring is given by:

$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$

Consider two events A and B with event B dependent on A. The probability of the occurrence of both events is given by:

$P[A \text{ and } B] = P[A] \times P[B \mid A]$

Probability Distributions:

$P[X = r] = {}^nC_r p^r q^{n-r} = \frac{n!}{(n-r)!r!}p^r q^{n-r}$ (Binomial distribution)

$P[X = r] = \frac{e^{-\mu}\mu^r}{r!}$, where $\mu = np$ (Poisson distribution)

$a = \frac{x - \mu}{\sigma}$ (Normal distribution)

Statistics:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$ (Dataset mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{n} x_i^{\,2} - n(\bar{x})^2}{n-1}$$ (Dataset standard deviation)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$$ (Grouped data mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i - 1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{M} f_i m_i^{\,2} - M(\bar{x})^2}{M-1}$$ (Grouped standard deviation)

Derivation:

First Principles: $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ **Power Rule:** $\frac{dy}{dx} = nx^{n-1}$

Product Rule: $\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$ **Quotient Rule:** $\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Integral: $\int ax^n dx = \frac{ax^{n+1}}{n+1} + c, \; n \neq -1$

Definite Integral:

$$\int_a^b f'(x)dx = f(x)\Big]_a^b = f(b) - f(a)$$

Table of Probabilities – P[Z > a], where Z is the standard normal variable.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000